Aaron Wong

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EDUCATION

Georgia Institute of Technology (GT) — **Atlanta, GA**
Bachelor of Science in Aerospace Engineering, Minor in CS + AI, GPA: 3.89 — *Expected May 2027*
Masters of Science in Aerospace Engineering — *Expected May 2028*

- **Activities:** Grand Challenges LLC; GTSF Allocation Committee/FLI; Pep/Concert Band; HKSA
- **Relevant Coursework:**Thermofluids, Jet/Rocket Propulsion, Aerodynamics, Computer Organization, Dynamics

SKILLS

- **Programming:** Python, Java, Matlab, C, C++ (Intermediate), Assembly, AI/ML: scikit-learn, Keras, Git
- **Design:** Autodesk Inventor, Fusion, Solidworks CAD (Intermediate), CATIA, FEA, CST/EMA, CFD, Amesim
- **Machine Shop:** Laser Cutters, 3D-Printer, Electrical/Soldering, Waterjet, Lathe, Mill
- **General:** Microsoft 360 Suite, Canva, Figma

WORK AND LEADERSHIP EXPERIENCE

Propulsion and Thermofluids Intern — **Savannah, GA**
Gulfstream Aerospace Corporation — *January 2026 - May 2026*

- Evaluated capabilities of high fidelity finite difference electromagnetic models simulating lightning strike effects on fuel autoignition probabilities vs testing data. Resulted in over $100,000 in long-term licensing savings.
- Develop contrail prediction model based on elevation, humidity, and temperature data allowing customers to target low-contrail probability flight times and provide additional ESG data for corporate customers.
- Support aeroperformance optimization of Rolls-Royce Pearl 700 engines via data processing and simulation.
- Use Amesim to model and conduct 1D thermofluid analysis and simulate system control logic on fuel delivery and ECS systems for FAA G400 certification. Successfully designed logic to operate within conditions.

Research Assistant — **Atlanta, GA**
Ben T. Zinn Combustion Laboratory — *December 2024-Present*

- Designed and machined high-pressure feed system for methane combustor; integrating fiber-optic diagnostics (GE collaboration) to measure combustion stability and determine product species concentration.
- Conducted over 20 tests on novel fiber optic diagnostic and mounting methods for high pressure combustion.
- Learned and performed doctorate level signal processing processes for pressure and mass fraction data.

Propulsion Testing and Ignition Responsible Engineer — **Atlanta, GA**
GT Supersonics — *May 2025-Present*

- Led structural and CAD design of EXACT test stand rated for FOS = 3 for 400 lbf; integrated modular rail, electronic, and fluid systems, enable rapid test stand assembly and accurate thrust measurement(<5% error).
- Developing a 4’ extension of EXACT test stands for afterburner and fuselage-engine configuration testing.
- Established test safety protocols and procedures and direct mission control operations during static tests.
- Designed glowplug torch igniter chamber and M10 to NPT fitting. Designed electrical system and sourced parts to handle 30 Amp. Modeled ignition conditions using thermodynamic analysis.

Propulsion Team Lead — **Atlanta, GA**
Propulsive Landers @ GT — *May 2024-August 2025*

- Modeled hybrid rocket chamber and nozzle thermodynamics in MATLAB to optimize thrust efficiency and propellant utilization. Defined valve/line sizing and compiled BOM for prototype test stand.
- Developed test procedures for cold-flow and hot-fire tests. Tested successful ignition at 40% max thrust.

PROJECTS

Using Machine Learning Models to Analyze the Aerodynamic Properties of Airfoils — *Summer 2023*

- Scraped and cleaned airfoil image and numerical data and developed ML models using Keras to predict lift/drag coefficients of airfoils as a faster alternative to CFD (achieved 60% accuracy).

HONORS & AWARDS

- **AIME Qualifier top 1% (2020–2023):** 4-time qualifier; awarded Distinction (Top 5%). Peak score: 8 (Top 1%)
- **Lean Six Sigma:** White Belt